

GPU · Inference · Batch

An Indian AI startup ran unoptimised for 6 months. ₹68 lakh/month. Two weeks of optimisation — caching, batching, smaller model, cheaper embeddings — dropped the bill to ₹15 lakh. Runway extended by 8 months.

What does YOUR monthly LLM bill say about your caching strategy?

WHAT'S INSIDE

- 01 Inference cost formula
- 02 5 cost levers
- 03 GPU types × ₹/hour (Mumbai)
- 04 Spot + reserved + serverless
- 05 Analogy

WHY THIS LESSON MATTERS

\$82,000/month → \$18,000. Two weeks. No user loss.

NUMBER 1

₹68 lakh

From real-world deployments.

NUMBER 2

₹15 lakh

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Calculate inference cost per query.

2

5-lever cost framework.

3

Pick GPU type by workload.

4

Design spot + reserved GPU strategy.

5

Cut existing bill 50%+.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

**Inference cost
formula**

Per query + monthly

02

5 cost levers

Key concept of this lesson.

03

**GPU types × ₹/hour
(Mumbai)**

Key concept of this lesson.

04

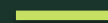
**Spot + reserved +
serverless**

Match commitment to workload

01

PART 1 OF 4

Inference cost formula



Per query + monthly

CONCEPT 1 OF 4

Inference cost formula

$\text{cost_per_query} = (\text{in} \times \text{p_in} + \text{out} \times \text{p_out}) / 1000$

$\text{monthly} = \text{cost_per_query} \times \text{queries_per_day} \times 30$

Output tokens 5-10× more expensive than input.

Ask for concise outputs first.

FIRST LEVER

Shorter outputs

Often cuts bill 30%+.

DEFINITION

Inference cost formula

Per query + monthly

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

5 cost levers

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

GPU types × ₹/hour (Mumbai)

A core idea you'll use throughout this lesson.

DEFINITION

GPU types × ₹/hour (Mumbai)

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Spot + reserved + serverless

Spot: up to 90% cheaper; reclaimable. Training.

Reserved 1-3yr: 30-70% off; steady prod.

Serverless GPU (Modal, Replicate): pay-per-call.

Mix: spot for training + reserved for prod baseline + on-demand spikes.

MIX

3 commitments

Match to workload; save 40-60%.

KEY POINTS

- Spot: up to 90% cheaper; reclaimable. Training
- Reserved 1-3yr: 30-70% off; steady prod
- Serverless GPU (Modal, Replicate): pay-per-call
- Mix: spot for training + reserved for prod baseline + on-demand spikes

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

5 levers:

A

Hardware

B

Cache+smaller+short+batch+structured

C

Only caching

D

Random

5 levers:

CORRECT ANSWER



Cache+smaller+short+batch+structured

WHY

The 5-lever framework

T4 best for:

A 70B

B Dev + small inference

C Frontier training

D Nothing

T4 best for:



CORRECT ANSWER

Dev + small inference

WHY

16 GB VRAM; cheap

Spot GPUs best for:

A

Interactive

B

Training with checkpoints

C

Latency-critical

D

DBs

Spot GPUs best for:



CORRECT ANSWER

Training with checkpoints

WHY

Up to 90% cheaper; interruptible

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

$\text{cost_per_query} = (\text{in} \times \text{p_in} + \text{out} \times \text{p_out}) / 1000$
 $\text{monthly} = \text{cost_per_query} \times \text{queries_per_day} \times 30$
Output tokens 5-10× more expensive than input.
Ask for concise outputs first.
FIRST LEVER
Shorter outputs
Often cuts bill 30%+.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

- 1 Inference cost formula
- 2 5 cost levers
- 3 GPU types × ₹/hour (Mumbai)
- 4 Spot + reserved + serverless

Output tokens cost more.

"Ask for less. Cache more. Batch when you can."

DELIVERABLES

- NEXT
- Projects Saturday — ship to prod.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.